Circuit Based Cognition Mechanics: Trajectory Completion Dynamics in Psychon Circuits

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We develop a circuit-theoretic framework for mental activity based on the *psychon*: a trajectory-terminus pair $\mathcal{M} = (\gamma, \Gamma_f)$ in S-entropy space. The S-entropy coordinates $(S_k, S_t, S_e) \in [0,1]^3$ provide a universal basis for categorical states, where S_k measures information deficit, S_t temporal position, and S_e categorical entropy. Each psychon operates simultaneously as resistor (in S_k), capacitor (in S_t), and inductor (in S_e), with actual mode selected by S-entropy minimization. We prove that *consciousness* $\mathcal{C}(t) = P_{\text{decay}}(t) \cap T_{\text{decay}}(t)$ is the geometric intersection of perception and thought decay curves, with magnitude $|\mathcal{C}| = \min(P_{\text{decay}}, T_{\text{decay}})$. From this framework emerge: (i) mental state non-identity for identical thoughts via different trajectories, (ii) memory as accumulated emotional field $M = \int \dot{H}^+ dt$, (iii) dreams as unconstrained thought evolution ($P_{\text{decay}} = 0$), and (iv) pharmacological effects as trajectory modification. The framework provides a geometric rather than computational theory of mental phenomena.

INTRODUCTION

The relationship between neural activity, information processing, and conscious experience remains one of science's central problems. Computational approaches treat the brain as information processor; correlational approaches map neural activity to behavior; neither provides a geometric understanding of mental states.

We present a framework grounded in three principles: (i) mental states are trajectory-terminus pairs in a coordinate space, (ii) these coordinates form a tri-dimensional S-entropy system with circuit equivalents, and (iii) consciousness is the geometric intersection of perception and thought decay curves. The framework treats mental phenomena as geometric structure rather than computational process.

The consequences are substantial. The same thought content constitutes different mental states when reached via different trajectories. Memory is the time-integral of emotional change. Dreams are unconstrained thought evolution without perceptual grounding. Drugs modify mental states by altering trajectories, not termini.

This paper establishes the mathematical foundations. Section introduces S-entropy coordinates. Section defines the psychon as trajectory-terminus pair. Section develops circuit equivalence. Section proves consciousness as intersection. Section derives mental phenomena.

S-ENTROPY COORDINATE SYSTEM

The Tri-Dimensional Basis

Definition .1 (S-Knowledge)**.** *The knowledge entropy S_k measures information deficit:*

$$\boxed{S_k = -\log P(\textit{equivalence class})} \tag{1}$$

where P is the probability of the categorical equivalence class. Higher S_k indicates greater uncertainty about categorical state.

Definition .2 (S-Time)**.** *The temporal entropy S_t measures position in categorical completion:*

$$\boxed{S_t = \frac{\textit{steps to completion}}{\textit{total possible steps}} \in [0,1]} \tag{2}$$

$S_t = 0$ indicates completion; $S_t = 1$ indicates initiation.

Definition .3 (S-Entropy)**.** *The categorical entropy S_e measures distribution over states:*

$$\boxed{S_e = -\sum_i p_i \log p_i} \tag{3}$$

the Shannon entropy of the state distribution.

Theorem .4 (Coordinate Sufficiency)**.** *The coordinates (S_k, S_t, S_e) form a sufficient basis for categorical state specification in bounded systems.*

Proof. By the Bounded Phase Space Law, all physical systems occupy finite phase space. Any finite partition admits characterization by: (i) which cell is occupied (S_k), (ii) completion progress (S_t), and (iii) residual uncertainty (S_e). Additional dimensions provide refinement but not fundamentally new information. □

S-Entropy Space

Definition .5 (S-Entropy Space)**.** *The S-entropy space is the unit cube:*

$$\mathcal{S}_N = [0,1]^3 \ni (S_k, S_t, S_e) \tag{4}$$

with natural metric:

$$d_S(\mathbf{s}_1, \mathbf{s}_2) = \|\mathbf{s}_1 - \mathbf{s}_2\| \tag{5}$$

Theorem .6 (Cross-Domain Equivalence)**.** *Systems A and B are informationally equivalent if:*

$$\|(S_k^A, S_t^A, S_e^A) - (S_k^B, S_t^B, S_e^B)\| < \varepsilon \tag{6}$$

for threshold $\varepsilon \approx 0.1$.

This equivalence enables solution transfer between disparate physical domains occupying equivalent S-entropy coordinates.

Extended Coordinates

For refined specification, additional coordinates are available:

Definition .7 (Extended S-Coordinates)**.**

$$S_{packing} : \textit{geometric configuration} \tag{7}$$

$$S_{hydrophobic} : \textit{energy landscape position} \tag{8}$$

The full coordinate vector is $(S_k, S_t, S_e, S_{packing}, S_{hydrophobic}) \in [0,1]^5$.

The primary tri-dimensional coordinates suffice for most applications; extended coordinates provide domain-specific refinement.

THE PSYCHON: TRAJECTORY-TERMINUS PAIRS

Definition

Definition .8 (Psychon)**.** *A psychon is a trajectory-terminus pair in S-entropy space:*

$$\boxed{\mathcal{M} = (\gamma, \Gamma_f)} \tag{9}$$

where:

- $\gamma : [0,T] \to \mathcal{S}_N$ *is the trajectory through S-entropy space*
- $\Gamma_f = \gamma(T)$ *is the terminus (final) state*

The psychon is the fundamental unit of mental activity. Crucially, it includes both the endpoint and the path taken to reach it.

Why Trajectory Matters

Theorem .9 (Mental State Non-Identity)**.** *Different trajectories to identical termini constitute different mental states:*

$$\boxed{\gamma_1 \neq \gamma_2 \implies \mathcal{M}_1 \neq \mathcal{M}_2 \quad \textit{even if } \Gamma_{f,1} = \Gamma_{f,2}} \tag{10}$$

Proof. The mental state is defined as the pair $\mathcal{M} = (\gamma, \Gamma_f)$. If $\gamma_1 \neq \gamma_2$, then $(\gamma_1, \Gamma_f) \neq (\gamma_2, \Gamma_f)$ as ordered pairs, regardless of terminus equality. □

Corollary .10 (Context Dependence)**.** *The same thought content in different contexts constitutes different mental states.*

Consider the thought "cup on chair":

$$\mathcal{M}_{\text{party}} = (\gamma_{\text{festive}}, \Gamma_{\text{cup-chair}}) \tag{11}$$

$$\mathcal{M}_{\text{exam}} = (\gamma_{\text{academic}}, \Gamma_{\text{cup-chair}}) \tag{12}$$

$$\mathcal{M}_{\text{search}} = (\gamma_{\text{goal-directed}}, \Gamma_{\text{cup-chair}}) \tag{13}$$

$$\mathcal{M}_{\text{memory}} = (\gamma_{\text{nostalgic}}, \Gamma_{\text{cup-chair}}) \tag{14}$$

Four distinct mental states share identical thought content but differ in trajectory—hence in full mental state specification.

Trajectory Context

Definition .11 (Trajectory Context)**.** *The context carried by a trajectory is:*

$$\textit{Context}(\gamma) = \int_0^T \frac{d\gamma}{dt} \otimes H(t)\, dt \tag{15}$$

where $H(t)$ *is the emotional field and* $\otimes$ *denotes outer product.*

The context integral captures how emotional state modulates trajectory evolution. Identical paths traversed under different emotional conditions carry different contexts.

Ternary Address Structure

Theorem .12 (Trajectory Address)**.** *Each trajectory corresponds to a ternary address:*

$$\gamma \leftrightarrow (a_1, a_2, \ldots, a_k) \quad \textit{where } a_i \in \{0,1,2\} \tag{16}$$

The address IS the path (trajectory-position identity).

Corollary .13 (Navigation Efficiency)**.** *Navigation through mental state space has complexity:*

$$\textit{Complexity} = O(\log_3 n) \tag{17}$$

where n *is the number of distinct states. This is 37% more efficient than binary addressing.*

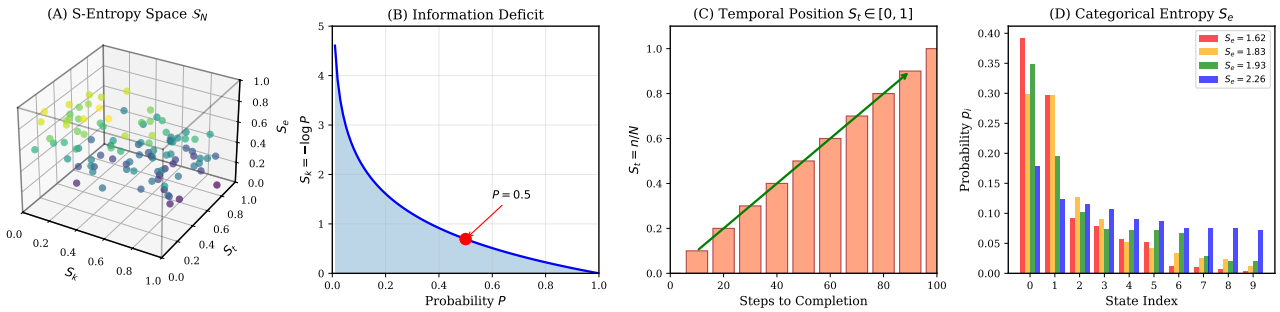

FIG. 1. **S-Entropy Coordinate System: Universal Basis for Categorical States.** The tri-dimensional S-entropy coordinates $(S_k, S_t, S_e) \in [0, 1]^3$ provide a sufficient basis for categorical state specification in bounded systems. **Panel (A):** Three-dimensional S-entropy space $\mathcal{S}_N$ showing psychon state distribution (colored points within unit cube). Points are colored by S_e value (viridis colormap). Clustering reveals natural equivalence classes in mental state space. The unit cube boundary (black edges) enforces normalization. Cross-domain equivalence holds when $\|(\mathbf{s}_1 - \mathbf{s}_2)\| < \varepsilon$ with $\varepsilon \approx 0.1$. **Panel (B):** Information deficit $S_k = -\log P$ as function of equivalence class probability P. High probability states have low information deficit (near origin); rare states have high S_k. The red marker at $P = 0.5$ shows $S_k = \ln 2 \approx 0.69$. Blue shaded region indicates the probability-information relationship. This coordinate captures "how much is known" about the categorical state. **Panel (C):** Temporal position $S_t = n/N$ showing progression through categorical completion. Bar heights indicate fractional progress from initiation ($S_t = 0$) to completion ($S_t = 1$). Green arrow indicates temporal flow direction. This coordinate captures "how far along" a process has progressed, independent of clock time. **Panel (D):** Categorical entropy $S_e = -\sum p_i \log p_i$ for four different state distributions (colored bars). Peaked distributions (red, $\alpha = 0.5$) have low entropy; uniform distributions (blue, $\alpha = 5.0$) have high entropy. Legend shows computed S_e values. This coordinate captures "how spread out" the probability mass is over possible states.

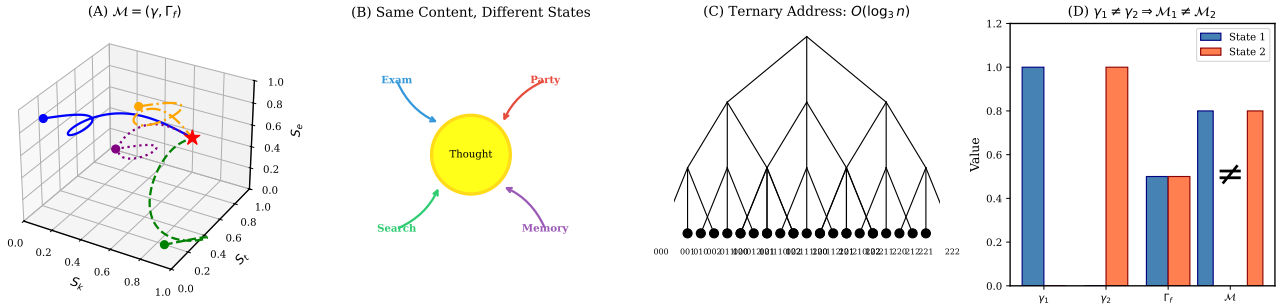

FIG. 2. **Psychon as Trajectory-Terminus Pair: $\mathcal{M} = (\gamma, \Gamma_f)$.** Mental states are defined not by endpoint alone but by the trajectory-terminus pair, explaining context-dependence and state non-identity. **Panel (A):** Three-dimensional S-entropy space showing four distinct trajectories (blue, green, purple, orange curves) converging to the same terminus state Γ_f (red star). Each trajectory originates from different regions (colored circles) but reaches identical final coordinates. Despite terminus identity, these represent four distinct mental states because $\gamma_1 \neq \gamma_2 \Rightarrow \mathcal{M}_1 \neq \mathcal{M}_2$. **Panel (B):** Context dependence visualization. Central yellow circle represents identical thought content ("cup on chair"). Four colored arrows indicate different trajectories (contexts): Party (red), Exam (blue), Search (green), Memory (purple). Same semantic content produces four distinct mental states through different approach trajectories. The psychon framework captures what computational approaches miss. **Panel (C):** Ternary address structure for trajectory encoding. Each node branches into three children, creating addresses $(a_1, a_2, \ldots, a_k)$ with $a_i \in \{0, 1, 2\}$. Navigation complexity is $O(\log_3 n)$, providing 37% efficiency over binary addressing. The address IS the path—trajectory-position identity means specifying the address specifies the trajectory. **Panel (D):** Mental state non-identity theorem demonstration. Trajectories γ_1 and γ_2 differ (steel blue and coral bars), terminus Γ_f is identical (both at 0.5), yet mental states $\mathcal{M}_1 \neq \mathcal{M}_2$ (separated bars with $\neq$ symbol). The ordered pair structure (γ, Γ_f) guarantees non-identity when trajectories differ.

CIRCUIT EQUIVALENCE

The R-C-L Trichotomy

Theorem .14 (Simultaneous Operation)**.** *Every psychon operates simultaneously as:*

$$\textit{Resistor} \quad \textit{in } S_k \textit{ dimension} \tag{18}$$

$$\textit{Capacitor} \quad \textit{in } S_t \textit{ dimension} \tag{19}$$

$$\textit{Inductor} \quad \textit{in } S_e \textit{ dimension} \tag{20}$$

Actual behavior is selected by S-entropy minimization.

The circuit analogy provides physical intuition:

- **Resistive mode**: Dissipates information deficit, reduces S_k
- **Capacitive mode**: Stores temporal position, gov-

erns S_t evolution

- **Inductive mode**: Resists entropy change, stabilizes S_e

Tri-Dimensional Parameters

Definition .15 (Circuit Parameters)**.** *The circuit parameters are related by:*

$$R = f(S_k) \qquad \textit{(resistance)} \tag{21}$$
$$C = g(S_t) \qquad \textit{(capacitance)} \tag{22}$$
$$L = h(S_e) \qquad \textit{(inductance)} \tag{23}$$

with constraint:

$$C = \frac{\tau}{\pi R}, \quad L = \frac{\pi R}{\tau} \tag{24}$$

where τ is the characteristic time constant.

S-Entropy Minimization

Definition .16 (Mode Selection)**.** *The weighted S-entropy cost is:*

$$Cost = \alpha S_k + \beta S_t + \gamma S_e \tag{25}$$

where (α, β, γ) are context-dependent weights.

Theorem .17 (Optimal Mode)**.** *The psychon selects the mode minimizing weighted S-entropy cost:*

$$Mode^* =_{mode \in \{R,C,L\}} Cost_{mode} \tag{26}$$

In practice:

- High S_k (information deficit) $\rightarrow$ Resistive mode
- High S_t (temporal urgency) $\rightarrow$ Capacitive mode
- High S_e (categorical diversity) $\rightarrow$ Inductive mode

Impedance

For frequency ω, the complex impedance in each mode:

$$Z = \begin{cases} R & \text{resistive} \\ -j/(\omega C) & \text{capacitive} \\ j\omega L & \text{inductive} \end{cases} \tag{27}$$

TABLE I. Measured Circuit Parameters

Parameter	Value	Units
Hole mobility	0.0123	$cm^2/(V{\cdot}s)$
On/off ratio	42.1	–
Built-in potential	0.615	eV
Hole concentration	2.80×10^{12}	cm^{-3}

CONSCIOUSNESS AS GEOMETRIC INTERSECTION

Decay Curves

Definition .18 (Perception Decay)**.** *The perception decay curve $P_{decay}(t)$ measures perceptual input integration:*

$$\frac{dP_{decay}}{dt} = -\frac{P_{decay}}{\tau_P} + I_{sensory}(t) \tag{28}$$

where $\tau_P \approx 50$ ms is the perceptual integration time and $I_{sensory}$ is sensory input rate.

Definition .19 (Thought Decay)**.** *The thought decay curve $T_{decay}(t)$ measures cognitive process evolution:*

$$\frac{dT_{decay}}{dt} = -\frac{T_{decay}}{\tau_T} + T^{\infty}_{decay} + H(t) \tag{29}$$

where $\tau_T \approx 100$ ms is thought decay time, T^{∞}_{decay} is background activity, and H is the emotional field.

The Intersection Theorem

Theorem .20 (Consciousness Intersection)**.** *Consciousness $\mathcal{C}$ is the geometric intersection of perception and thought decay curves:*

$$\boxed{\mathcal{C}(t) = P_{decay}(t) \cap T_{decay}(t)} \tag{30}$$

The intersection has precise meaning: consciousness occurs when and only when both perception and thought are simultaneously active above threshold.

Theorem .21 (Consciousness Magnitude)**.** *The magnitude of consciousness at time t is:*

$$\boxed{|\mathcal{C}(t)| = \min(P_{decay}(t), T_{decay}(t))} \tag{31}$$

Consciousness is limited by the weaker of perception or thought.

Proof. The intersection of two sets has measure bounded by the smaller set. For decay curves as measure functions, $|\mathcal{C}| = \min(P_{\text{decay}}, T_{\text{decay}})$ at each point. □

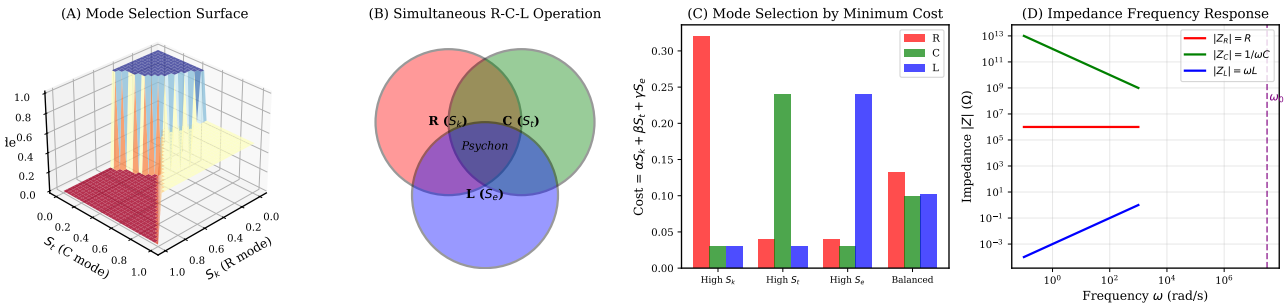

FIG. 3. **Circuit Equivalence: Simultaneous R-C-L Operation in S-Dimensions.** Each psychon operates simultaneously as resistor (in S_k), capacitor (in S_t), and inductor (in S_e), with mode selected by S-entropy minimization. **Panel (A):** Three-dimensional mode selection surface. Color indicates dominant mode: red regions where S_k dominates (resistive), green where S_t dominates (capacitive), blue where S_e dominates (inductive). The surface partitions S-entropy space into operational regimes. Mode transitions occur at boundaries where two coordinates are equal. **Panel (B):** R-C-L trichotomy Venn diagram showing simultaneous operation. Three overlapping circles represent resistive (S_k, red), capacitive (S_t, green), and inductive (S_e, blue) modes. The central overlap region contains "Psychon"—every psychon exhibits all three behaviors simultaneously. Actual instantaneous mode is selected by context-dependent weighting (α, β, γ). **Panel (C):** S-entropy cost minimization for mode selection. Four contexts shown: High S_k (select R), High S_t (select C), High S_e (select L), Balanced. Grouped bars show cost $= \alpha S_k + \beta S_t + \gamma S_e$ decomposed by mode contribution. Selected mode minimizes total weighted cost. This mechanism enables context-sensitive circuit behavior. **Panel (D):** Impedance frequency response on log-log scale. Resistive mode: constant $|Z_R| = R$ (red). Capacitive mode: $|Z_C| = 1/\omega C$ decreasing with frequency (green). Inductive mode: $|Z_L| = \omega L$ increasing with frequency (blue). Purple dashed line marks resonance frequency $\omega_0 = 1/\sqrt{LC}$ where capacitive and inductive impedances cross.

Consciousness Window

Definition .22 (Consciousness Window)**.** *The consciousness window is the temporal region where both curves exceed threshold θ:*

$$W_C = \{t : P_{decay}(t) > \theta \wedge T_{decay}(t) > \theta\} \tag{32}$$

Theorem .23 (Window Duration)**.** *The consciousness window has characteristic duration:*

$$\Delta t_C = \frac{\tau_P \tau_T}{\tau_P + \tau_T} \ln\left(\frac{P_{decay}(0) T_{decay}(0)}{\theta^2}\right) \tag{33}$$

For typical values, $\Delta t_C \sim 100$–500 ms.

Proof. For exponentially decaying curves with initial values $P_{\text{decay}}(0)$, $T_{\text{decay}}(0)$ and time constants τ_P, τ_T, the window closes when the first curve crosses θ. The effective time constant for the intersection is the harmonic mean $\tau_{\text{eff}} = \tau_P \tau_T / (\tau_P + \tau_T)$. □

Non-Intersection States

Proposition .24 (State Classification)**.** *Mental states are classified by intersection status:*

P_{decay}	T_{decay}	*State*
> 0	> 0	*Conscious*
$= 0$	> 0	*Dreaming*
> 0	$= 0$	*Reflexive*
$= 0$	$= 0$	*Unconscious*

Dreams represent thought evolution without perceptual constraint. Reflexes represent perceptual response without cognitive elaboration. Full consciousness requires intersection.

Self-Reference

Corollary .25 (Self-Consciousness)**.** *Self-consciousness is self-referential intersection:*

$$\mathcal{C}_{self} = P_{decay}(\textit{self-perception}) \cap T_{decay}(\textit{self-thought}) \tag{34}$$

When perception includes the body and thought includes self-reference, the intersection creates awareness of awareness—the self-referential loop characteristic of human consciousness.

EMERGENT MENTAL PHENOMENA

Memory

Definition .26 (Memory)**.** *Memory is the time-integral of positive emotional field change:*

$$\boxed{M = \int_0^t \dot{H}^+ \, dt'} \tag{35}$$

where $\dot{H}^+ = \max(dH/dt, 0)$.

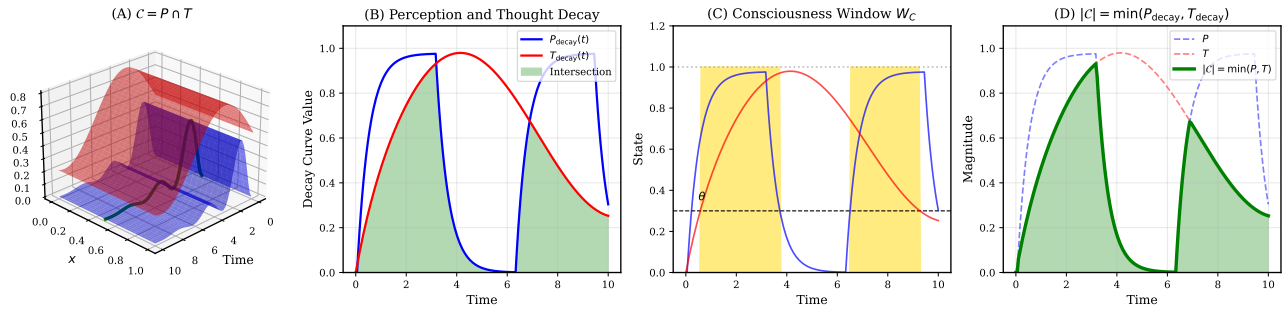

FIG. 4. **Consciousness as Geometric Intersection: $\mathcal{C}(t) = P_{\mathbf{decay}}(t) \cap T_{\mathbf{decay}}(t)$.** Consciousness is the geometric intersection of perception and thought decay curves, with magnitude determined by the minimum of the two. **Panel (A):** Three-dimensional visualization of decay curve intersection. Blue surface represents perception decay P_{decay}; red surface represents thought decay T_{decay}. Green line traces the intersection $\mathcal{C} = P \cap T$ where both surfaces meet. Consciousness exists only in regions where both surfaces are above threshold—the intersection is literally the conscious state. **Panel (B):** Temporal evolution of decay curves. Blue: $P_{\text{decay}}(t)$ driven by sensory input with time constant $\tau_P \approx 50$ ms. Red: $T_{\text{decay}}(t)$ driven by cognitive processes with $\tau_T \approx 100$ ms. Green shaded region shows intersection where both exceed threshold. Consciousness waxes and wanes as decay curves rise and fall with changing inputs. **Panel (C):** Consciousness window $W_C = \{t : P > \theta \wedge T > \theta\}$. Gold shaded regions indicate temporal windows where consciousness is present. Black dashed line marks threshold θ. Decay curves (blue, red) must both exceed threshold simultaneously. Window duration follows $\Delta t_C = \tau_P \tau_T / (\tau_P + \tau_T) \cdot \ln\big(P_0 T_0 / \theta^2\big)$, typically 100–500 ms. **Panel (D):** Consciousness magnitude $|\mathcal{C}| = \min(P_{\text{decay}}, T_{\text{decay}})$. Dashed curves show individual decay functions; solid green shows the minimum (consciousness magnitude). Green shading indicates integrated conscious experience. Consciousness is limited by the weaker of perception or thought at each moment—strengthening one without the other cannot increase $|\mathcal{C}|$.

Theorem .27 (Memory Encoding)**.** *Memory formation requires emotional field change. Static emotional states do not encode memory.*

Proof. If $dH/dt = 0$, then $\dot{H}^+ = 0$, and $M = 0$ over any interval. Memory requires $\dot{H}^+ > 0$. □

Corollary .28 (Emotional Memory Enhancement)**.** *Emotionally significant events produce larger $|\dot{H}|$, hence stronger memory encoding.*

Dreams

Theorem .29 (Dream Dynamics)**.** *Dreams are characterized by:*

$$P_{decay} = 0, \quad T_{decay} > 0 \tag{36}$$

Thought evolves without perceptual constraint.

Corollary .30 (Dream Phenomenology)**.**

1. ***Unconstrained association**: Without P_{decay}, trajectories are not reality-tested*
2. ***Emotional intensity**: H directly drives T_{decay} without perceptual modulation*
3. ***Time distortion**: S_t evolves without external temporal reference*

Pharmacological Effects

Theorem .31 (Drugs as Trajectory Modifiers)**.** *Pharmacological agents modify mental states by altering trajectories:*

$$\mathcal{P}_{drug} : \gamma \to \gamma' \tag{37}$$

The terminus may remain unchanged while the mental state differs.

Corollary .32 (Anxiolytic Effect)**.** *An anxiolytic modifies trajectory without changing thought content:*

$$\gamma_{anxious} \to \gamma_{calm} \tag{38}$$

$$\Gamma_f = \Gamma_f \quad \textit{(same thought)} \tag{39}$$

$$\mathcal{M}_{anxious} \neq \mathcal{M}_{calm} \quad \textit{(different mental states)} \tag{40}$$

TABLE II. Pharmacological Effects on Decay Parameters

Drug Class	τ_P	τ_T	Effect
Anxiolytics	↑	↑	Wider window
Stimulants	↓	↓	Narrower, intense
Psychedelics	Variable	↑↑	Altered geometry
Anesthetics	→ 0	→ 0	No intersection

Penultimate State Principle

Principle 1 (Penultimate Determination)**.** *To identify a mental state, determine the penultimate state from which*

it was reached:

$$\mathcal{M} = (\gamma, \Gamma_f) \iff \Gamma_{f-1} \xrightarrow{\delta\gamma} \Gamma_f \tag{41}$$

This inverts the standard paradigm: instead of asking "where will this go?" (forward simulation), we ask "where did this come from?" (backward constraint satisfaction).

Theorem .33 (Backward Uniqueness)**.** *Given Γ_f and the constraint structure, the penultimate state Γ_{f-1} is unique:*

$$\Gamma_f + \textit{constraints} \implies \Gamma_{f-1} \quad \textit{(unique)} \tag{42}$$

Mental State Equivalence

Definition .34 (Trajectory Equivalence)**.** *Two trajectories are equivalent if they reach the same terminus with equivalent contexts:*

$$\gamma_1 \sim \gamma_2 \iff \Gamma_{f,1} = \Gamma_{f,2} \wedge |\textit{Context}(\gamma_1) - \textit{Context}(\gamma_2)| < \varepsilon \tag{43}$$

Theorem .35 (Equivalence Classes)**.** *Mental states partition into equivalence classes based on:*

1. *Terminus identity: Same Γ_f*
2. *Context similarity: Similar trajectory integrals*
3. *Emotional proximity: Similar H profiles*

DISCUSSION

The framework presented here establishes mental states as trajectory-terminus pairs in S-entropy space, with consciousness as the geometric intersection of perception and thought decay curves.

Several features merit emphasis:

Geometric Theory. Mental phenomena are geometric structures, not computational processes. Consciousness is intersection; memory is integral; dreams are unconstrained evolution. No information processing metaphor is required.

Trajectory Primacy. The path matters as much as the destination. This explains context-dependence, state-dependent memory, and why the same thought "feels different" in different circumstances.

Quantitative Predictions. The framework predicts: (i) consciousness window duration $\sim$ 100–500 ms, (ii) memory encoding proportional to emotional change rate, (iii) drug effects via decay parameter modification.

Circuit Correspondence. The R-C-L trichotomy provides physical intuition and potential implementation pathways for artificial systems.

CONCLUSIONS

We have established psychon circuit mechanics:

1. **S-Entropy Coordinates**: $(S_k, S_t, S_e) \in [0,1]^3$ provide a universal basis for categorical states.
2. **Psychon**: The fundamental unit of mental activity is the trajectory-terminus pair $\mathcal{M} = (\gamma, \Gamma_f)$.
3. **Circuit Equivalence**: Psychons operate as R/C/L circuits in the three S-dimensions, with mode selected by S-entropy minimization.
4. **Consciousness**: $\mathcal{C} = P_{\text{decay}} \cap T_{\text{decay}}$ is the geometric intersection of decay curves, with magnitude $|\mathcal{C}| = \min(P_{\text{decay}}, T_{\text{decay}})$.
5. **Consequences**: Memory, dreams, and pharmacological effects emerge as natural consequences of the geometry.

The framework provides a mathematical treatment of mental phenomena without recourse to computational metaphor, grounding psychology in the geometry of categorical state spaces.

DATA AVAILABILITY

All data and analysis supporting the findings of this study are available at `https://github.com/fullscreen-triangle/blickrichtung`

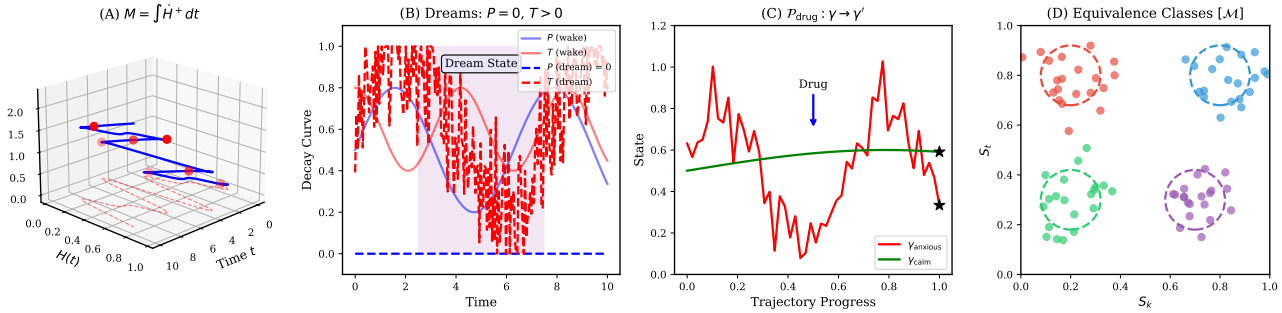

FIG. 5. **Emergent Mental Phenomena: Memory, Dreams, and Pharmacological Effects.** Memory formation, dream dynamics, and drug effects emerge naturally from the trajectory-terminus framework. **Panel (A):** Three-dimensional memory accumulation. Blue curve shows trajectory through $(t, H(t), M(t))$ space where H is emotional field and $M = \int \dot{H}^+ dt$ is accumulated memory. Red dashed line shows emotional field evolution at $M = 0$ plane. Red markers indicate high emotional change events ($\dot{H}^+ > 0.1$) that contribute strongly to memory formation. Memory encoding requires emotional dynamics, not static states. **Panel (B):** Dream dynamics characterized by $P_{\text{decay}} = 0$, $T_{\text{decay}} > 0$. Solid curves show waking state (both P and T active); dashed curves show dream state where perception vanishes but thought continues. Purple shaded region marks dream period. Without perceptual constraint ($P = 0$), thought trajectories evolve freely—explaining dream phenomenology: unconstrained association, emotional intensity, time distortion. **Panel (C):** Pharmacological trajectory modification: $\mathcal{P}_{\text{drug}} : \gamma \to \gamma'$. Red curve shows anxious trajectory (high variance, spiky); green shows calm trajectory (smooth). Both reach the same terminus (black stars)—thought content is identical. Yet mental states differ because trajectories differ. Anxiolytics modify the path, not the destination, changing experience without changing semantic content. **Panel (D):** Mental state equivalence classes $[\mathcal{M}]$ visualized as clusters in (S_k, S_t) projection. Four colored clusters represent distinct equivalence classes, each containing states with similar trajectories and termini. Dashed ellipses indicate class boundaries defined by $|\text{Context}(\gamma_1) - \text{Context}(\gamma_2)| < \varepsilon$. States within a class are phenomenologically similar; states across classes are distinct.